

12. Analysis C: true vs sensed area

Sensed area is compared with two references. One is the true area of the as-built feature. The other is the area that stereo geometry allows the sensor to see. The second matters for cutouts: part of the back plate seen through a hole is visible to only one camera, or is not lit by the projector, so no stereo sensor could read it.

1. Feature window. For each feature, frame, and pose, project the as-built outline into the image using the registered pose. The window is a circle of radius $D/2$ plus the 8 px analysis band (`BOUNDARY_BAND_HALF_WIDTH_PX`), in px at that Z , around the projected center. A feature whose window does not lie entirely inside the image is skipped and counted in the details.
2. Reference planes. Fit the front plane to an annulus of front surface outside the window. For a cutout this is the front plate. For a disk it is the registered disk plane, because small disk faces have too few pixels to fit. Fit the back plane to an annulus of back plate beyond the shadow band.
3. Pixel classes. Compute h as in 11.1. Inside the window, each pixel is a front read (h at or above 0.5, `FRONT_READ_HEIGHT_THRESHOLD`), a back read (h below 0.5), or a no-read.
4. Sensed area per frame. For a disk, sensed pixels are the front reads connected to the projected center. For a cutout, they are the back reads connected to it. Each pixel's area on the feature plane is $a = Z_{\text{plane}}(u, v)^2 / (f_x f_y \cos \phi)$, where ϕ is the incidence angle. Report three quantities: $A_{\text{sensed}} = \sum a$ over sensed pixels; $A_{\text{noread}} = \sum a$ over no-reads inside the projected outline grown by one pixel; and $A_{\text{sensed}} + A_{\text{noread}}$, an upper bound for a reader that treats no-reads as part of the feature.
5. Sub-pixel area. For each pose, average its frames and interpolate h over valid pixels. Extract the iso-contour at $h = 0.5$ (`ESF_HALF_HEIGHT`) with marching squares, deproject it onto the feature plane, and take the polygon area. Use this to check the per-frame pixel-count result for systematic error.
6. True area. $A_{\text{true}} = \pi D^2/4$, using the as-built D .
7. Geometric visible area. The depth of a point can be sensed only where both IR cameras, at 0 and at the baseline B , see it and the laser projector lights it. Ray-cast from the registered left camera, right camera, and projector centers through the as-built outline to the back plane; A_{geo} for a cutout is the overlap of the three projected outlines. With the projector midway between the cameras, as the VSX3000's layout places it, every point of a convex outline that both cameras see is also lit, so the three-center overlap equals the two-camera one; compute both anyway with the as-built projector position (`PROJECTOR_OFFSET_MM`), and where they differ report which

tracks the data better. This is the visibility of the depth sense. The visibility of the 2-D image is computed the same way with a single center at the 2-D sensor's as-built position on the axis between the IR cameras (CAMERA_2D_OFFSET_MM) and its collocated illumination, so that a point is seen and lit along the same ray: A_geo_2D is the one-center projection of the outline, reported beside A_geo for reference. Whether a visible point is also bright enough to read follows from the light's properties and the camera, light, and surface geometry (the irradiance falls with the cosine of the incidence angle over the squared distance, scaled by the surface reflectance); the procedure records the illumination settings and the surface finish but does not characterize the 2-D image. For a disk front face, A_geo = A_true.

8. Transfer curves. Plot $A_{\text{sensed}} / A_{\text{true}}$ and $A_{\text{sensed}} / A_{\text{geo}}$ against $D_{\text{px}} = D \cdot f_x / Z$, pooled over the features and stations of the Z sweep with the feature identity kept as a factor, for each gap and feature type. Put disks and cutouts on the same axes so their asymmetry shows. Report the mean, the spread across the 30 jitter poses (PHASE_JITTER_POSES_AREA), which is the phase effect, and the spread within a pose over its 10 frames (FRAMES_PER_AREA_POSE), which is the temporal effect.
9. Scaling test. Where two neighboring features overlap in D_{px} (half an octave at each junction of the ladder), compare their transfer curves over the shared range: the statistic is the mean difference of the two curves over that range, and the curves agree when zero lies inside its bootstrap interval. One row per feature pair, kind, gap, and ratio goes to C_overlap_test.csv. Agreement supports D_{px} as the governing variable. Disagreement is attributed to the Z-dependent noise $\sigma_{\text{tot}}(Z)$ of Section 10 and is reported as the difference between the two curves.
10. Edge bias. Convert to an equivalent diameter $D_s = 2\sqrt{(A_{\text{sensed}}/\pi)}$. The edge bias is $b = (D_s - D)/2$, in mm and in px. Fit b against Z using the largest feature at the stations where its D_{px} is at least 14 px (AREA_BIAS_FIT_MIN_D_PX), twice the 7 px expected minimum detectable diameter of Section 3.2, so that b does not depend on D .
11. Consistency with B. Build a 2-D blur kernel from the H and V line spread functions that Section 11, Step 16 outputs (assumed separable); where they are not available, use a Gaussian whose 10–90% rise equals the rise distance and say so. Blur an ideal disk of the as-built D with it and threshold at 0.5; this predicts the area. Make the prediction twice: with the blur alone, and with the edge offset s_{50} of 11.1 applied as a growth of the front material, because a symmetric blur cannot shift the half-height crossing by itself. Agreement for large D and disagreement for small D means the sensor's interpolation fills in small features beyond what the edge spread function explains. Also check the sign of b against s_{50} : the front material grows by $-s_{50}$, so b for a disk should be about $-s_{50}$ and b for a cutout about $+s_{50}$.
12. Field and open-background variants. Compare the off-axis transfer curves with the on-axis ones, with the achieved field fraction of each pose (Section 7) as a factor. For the open-background

cutouts at the reference station, compare the no-read area with A_{true} : fill-in by front-plate values shows as $A_{\text{noread}} < A_{\text{true}}$.

13. Outputs. `C_area_summary.csv` with one row per feature, station, gap, field position, and sub-series; `C_overlap_test.csv`; `C_area_details.json`. Figures (PNG and SVG): the transfer curves with the overlap ranges shaded; b against Z ; phase spread against D_{px} ; predicted against measured area. No term of C enters `forward_model_parameters.json`; b is a cross-check of the boundary widths of Section 14.